

Week 5 Interim Submission

AI-Assisted Triage: Data Exploration (Part 1 of 2)

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Feasibility Verdict

This dataset is suitable for building a baseline triage model, provided that leakage columns are excluded, temperature is handled in Fahrenheit, and findings are not assumed to generalise to a Caribbean ED without local validation.

Dataset Summary

The Yale EMLC triage dataset contains 55,121 patient encounters and 226 columns, drawn from a US academic hospital ED. Each encounter includes patient demographics, seven triage vital signs, and 200 binary chief-complaint flags. The target variable is the Emergency Severity Index (ESI), a five-level acuity scale where ESI 1 is the most critically ill. No missing values were found in any column, which is unusual for a clinical dataset of this size and suggests pre-processing was applied upstream.

The patient caseload is adult-only (age 18–107, median 55). Most encounters are mid-acuity: ESI 3 accounts for 49% of cases (27,010 encounters), ESI 2 for 33% (17,924), and ESI 1 for just 77 encounters (0.1%). This severe class imbalance of very few critically ill patients relative to moderate-acuity patients means a model that ignores ESI 1 entirely would still achieve high overall accuracy. Special care must be taken to ensure the model catches the sickest patients, not just the most common ones.

Initial Exploration Notes

Missingness

No missing values were detected across all 226 columns. While this is a quality strength, it warrants scrutiny: clinical datasets this complete often have upstream imputation applied, or values recorded as zero rather than blank. This assumption should be documented.

Leakage risk

Two columns `disposition` (admitted vs. discharged) and `previousdispo` (previous visit outcome) are known only after the ED visit is complete. If used as model inputs, they would give the algorithm information it could not have at triage time, inflating performance metrics artificially. Both will be excluded from modelling.

Candidate data quality concerns

- Temperature is recorded in degrees Fahrenheit (mean 98.1°F, range 91.5–106°F). Clinical thresholds such as fever ($>38^{\circ}\text{C}$ / 100.4°F) and hypothermia ($<35^{\circ}\text{C}$ / 95°F) differ from Celsius-based guidelines. This must be accounted for in feature engineering.
- Extreme glucose outliers exist: 164 encounters record glucose above 600 mg/dL, with a maximum of 1,066 mg/dL. These values are physiologically possible in diabetic emergencies but may also represent data entry errors. They will be reviewed against ESI level before a decision to retain or cap is made.

- Chief-complaint flags are sparse binary columns 200 in total, but only 15 have more than 1,000 positive cases. Many columns will add minimal signal and may increase model noise.

Data Quality Dashboard (Draft)

The eight-panel dashboard below summarises the key data quality and clinical validity findings from the initial exploration. Full visualisation committed to carisurg-portfolio/docs/.

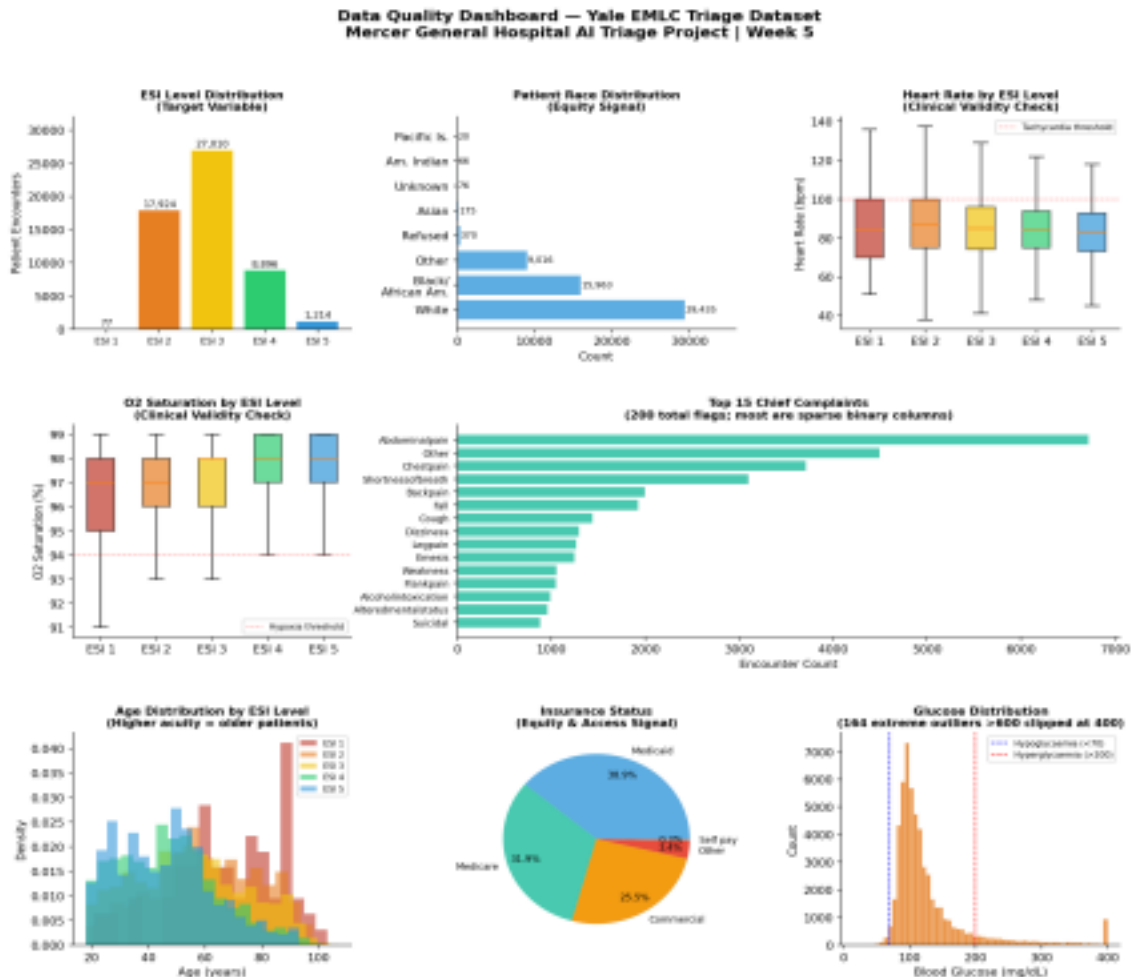


Figure 1 (draft). Data quality dashboard Yale EMLC triage dataset, 55,121 encounters.